Airfoil Project (Vortex Panel Method)

MAE 5500: Aerodynamics

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Problem 1: Vortex Panel Method Code Development for NACA 2412 with 10 Nodes

Write a code that uses the vortex-panel method outlined in section 1.6 to compute the lift coefficient, leading-edge pitching-moment coefficient, and quarter-chord pitching-moment coefficient at a specified angle of attack on an airfoil defined by a set of points that are read in from a text file. Use the points as the nodes in the algorithm.

For debugging purposes, the A matrix, B vector, and Gamma vector solutions for a NACA 2412 airfoil with 10 nodes at zero angle of attack can be viewed here. Results for this example should give the following:

$$\tilde{C}_L = 0.239837889916$$
$$\tilde{C}_{m_{LE}} = -0.111898723302$$
$$\tilde{C}_{m_{c/4}} = -0.0519392508233$$

The results for this problem are based on the Python script provided in my submission on Canvas. My results match the provided results in the NACA 2412 airfoil data above.

Problem 2: Creating VPM Table for NACA 2412 with 200 Nodes

Use the code developed for A1 to fill out the following table for the NACA 2412 airfoil with 200 nodes to 5 decimal places. The input to the code should include the following:

```
{
    "geometry": "2412_200.txt",
    "freestream_velocity" : 10.0,
    "alpha[deg]" : 0.0
}
```

Table 1: Vortex Panel Method Table Layout.

α [deg]	$\tilde{C}_L$	$\tilde{C}_{m_{LE}}$	$\tilde{C}_{m_{c/4}}$
-12			
-10			
-8			
-6			
-4			
-2			
0	0.25933	-0.12020	-0.05537
2			
4			
6			
8			
10			
12			
14			
16			

The following table is the collected results for $\tilde{C}_L$, $\tilde{C}_{m_{LE}}$, and $\tilde{C}_{m_{c/4}}$ at a given α value for the NACA 2412 airfoil.

Table 2: Vortex Panel Method data for NACA 2412 with 200 nodes.

α [deg]	$\tilde{C}_L$	$\tilde{C}_{m_{LE}}$	$\tilde{C}_{m_{c/4}}$
-12	-1.18434	0.24913	-0.04048
-10	-0.94564	0.19014	-0.04268
-8	-0.70578	0.12973	-0.04500
-6	-0.46505	0.06819	-0.04744
-4	-0.22377	0.00581	-0.04999
-2	0.01779	-0.05709	-0.05264
0	0.25933	-0.12020	-0.05537
2	0.50055	-0.18323	-0.05817
4	0.74116	-0.24587	-0.06103
6	0.98087	-0.30780	-0.06393
8	1.21939	-0.36874	-0.06686
10	1.45642	-0.42837	-0.06980
12	1.69167	-0.48641	-0.07274
14	1.92486	-0.54259	-0.07566
16	2.15571	-0.59661	-0.07856

Problem 3: $\tilde{C}_L$ Comparison Between VPM, TAT, and Experimental Data with 200 Nodes

Plot your lift-coefficient solution from A2 in comparison to predictions from thin-airfoil theory and experimental data. The zero-lift angle of attack from thin-airfoil theory is -2.077 degrees.

> Experimental data can be obtained from these tables. Use the data at a Reynolds number of about 3 million. What do you notice about the results?

Figure (1) compares the lift coefficients obtained from the Vortex Panel Method (VPM), Thin Airfoil Theory (TAT), and experimental data for the NACA 2412 airfoil.

The results show that all three methods produce similar lift coefficients for angles of attack between approximately $-12^\circ \leq \alpha \leq 4^\circ$. As the angle of attack increases beyond 4°, the results begin to diverge. Both VPM and TAT continue to predict a nearly linear increase in lift coefficient, while the experimental data begins to deviate from this behavior. This difference occurs because VPM and TAT assume inviscid flow and therefore do not account for viscous effects or boundary-layer separation.

As the angle of attack increases, the increasing pressure along the upper surface of the airfoil can cause the boundary layer to separate from the surface. This separation becomes more significant at higher angles of attack, eventually resulting in stall. The experimental data reaches a maximum lift coefficient of approximately 1.55 near $\alpha = 15^\circ$, after which the lift coefficient begins to decrease. In contrast, VPM and TAT do not predict stall because neither method accounts for viscous flow separation.

Another important observation is that the NACA 2412 airfoil produces positive lift at zero angle of attack, with a zero-lift angle of approximately -2°. This behavior is expected because the NACA 2412 is a cambered airfoil, allowing it to generate lift at zero angle of attack. Additionally, VPM generally predicts slightly higher lift coefficients than TAT at larger positive angles of attack. This difference can be attributed to VPM accounting for the actual airfoil geometry, including its thickness and camber, whereas TAT assumes an infinitesimally thin airfoil.

Overall, the results demonstrate that both theoretical methods provide reasonable lift predictions at lower angles of attack but become less accurate as viscous effects and flow separation become more significant.

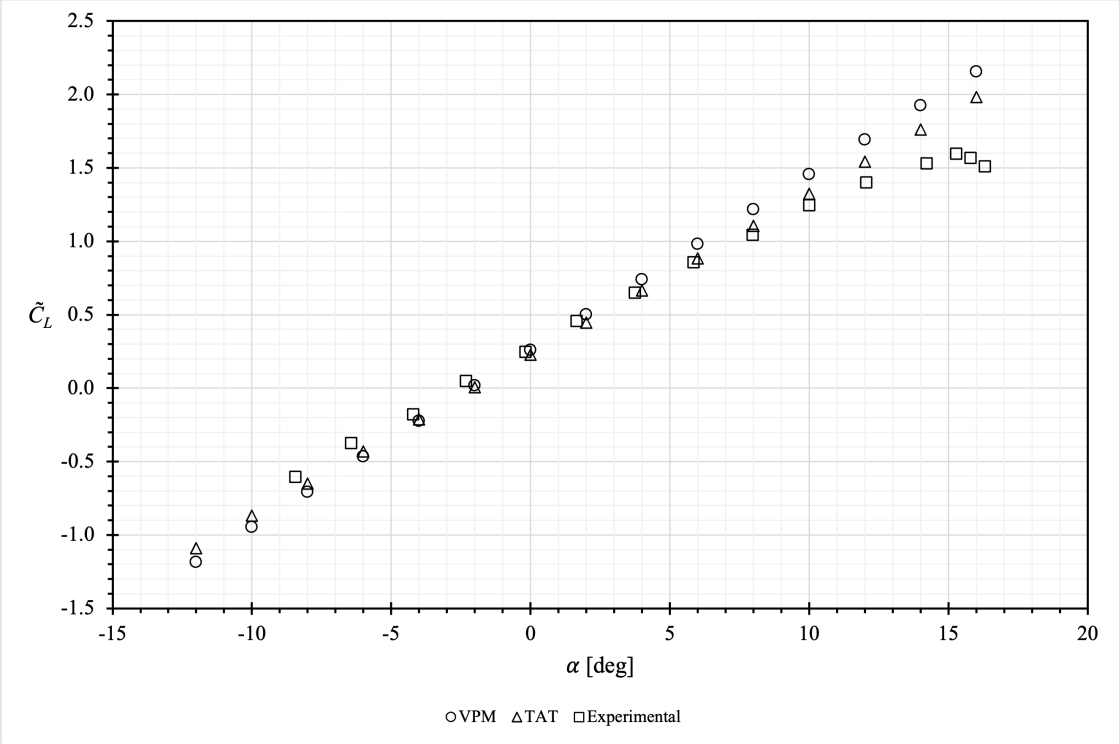


Figure 1: $\tilde{C}_L$ comparison between VPM, TAT, and Experimental Data for NACA 2412.

Problem 4: Same as Problem 3 Using NACA 2421 with 200 Nodes

Use your airfoil code to fill out the table shown in A2 for a NACA 2421 airfoil with 200 nodes. Plot your lift-coefficient results in comparison to predictions from thin-airfoil theory and experimental data. The zero-lift angle of attack from thin-airfoil theory is -2.077 degrees. Experimental data can be obtained from these tables. Use the data at a Reynolds number of about 3 million. What do you notice about the results?

Figure (2) compares the lift coefficients obtained from the Vortex Panel Method (VPM), Thin Airfoil Theory (TAT), and experimental data for the NACA 2421 airfoil.

The results show that all three methods produce similar lift coefficients at lower angles of attack, particularly between approximately $-6° \leq \alpha \leq 4°$. As the angle of attack increases beyond 4°, the results begin to diverge. VPM predicts slightly higher lift coefficients than TAT, while the experimental data increases at a slower rate. This difference becomes more significant at higher angles of attack because VPM and TAT assume inviscid flow and do not account for viscous effects or boundary-layer separation.

As the angle of attack increases, the increasing pressure along the upper surface of the airfoil can cause the boundary layer to separate from the surface. This separation reduces the rate at which lift increases and can eventually cause stall. The experimental data reaches a maximum lift coefficient of approximately 1.2 near $\alpha = 14°$, after which the lift coefficient begins to decrease slightly. In contrast, VPM and TAT continue predicting a nearly linear increase in lift coefficient because neither method accounts for viscous flow separation.

Another important observation is that the NACA 2421 airfoil has approximately the same zero-

lift angle of attack as the NACA 2412, near $-2°$. This is expected because both airfoils have the same camber and camber location, allowing them to produce positive lift at zero angle of attack. However, the NACA 2421 has a greater thickness, which influences the lift predictions. The difference between VPM and TAT becomes more noticeable for the thicker NACA 2421 airfoil because VPM accounts for the actual airfoil geometry, whereas TAT assumes an infinitesimally thin airfoil.

Overall, the results demonstrate that both theoretical methods provide reasonable lift predictions at lower angles of attack but become less accurate as the angle of attack increases. Comparing the NACA 2412 and NACA 2421 also demonstrates how increasing airfoil thickness can influence the lift characteristics, even when the camber and zero-lift angle remain approximately the same.

Table 3: Vortex Panel Method data for NACA 2421 with 200 nodes.

α [deg]	$\tilde{C}_L$	$\tilde{C}_{m_{LE}}$	$\tilde{C}_{m_{c/4}}$
-12	-1.25946	0.28411	-0.02388
-10	-1.00407	0.21840	-0.02880
-8	-0.74745	0.15108	-0.03397
-6	-0.48992	0.08246	-0.03935
-4	-0.23179	0.01289	-0.04492
-2	0.02662	-0.05731	-0.05065
0	0.28500	-0.12777	-0.05652
2	0.54303	-0.19817	-0.06250
4	0.80040	-0.26817	-0.06855
6	1.05679	-0.33740	-0.07466
8	1.31189	-0.40556	-0.08077
10	1.56540	-0.47229	-0.08688
12	1.81700	-0.53727	-0.09295
14	2.06639	-0.60019	-0.09894
16	2.31326	-0.66075	-0.10484

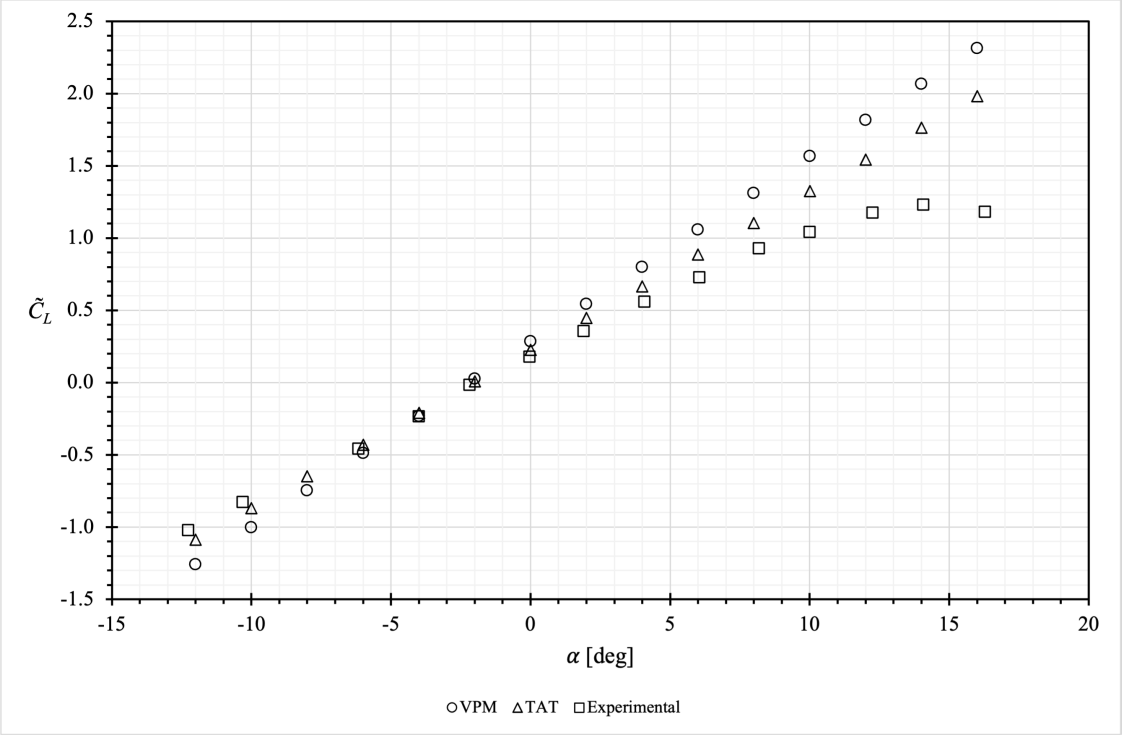


Figure 2: $\tilde{C}_L$ comparison between VPM, TAT, and Experimental Data for NACA 2421.

Problem 5: Same as Problem 3 Using NACA 0015 with 200 Nodes and XFOIL as the Experimental Data

Use your airfoil code to fill out the table shown in A2 for a NACA 0015 airfoil with 200 nodes. Use XFOIL to fill out the same table for the same airfoil at a Reynolds number of 3 million. Plot the lift-coefficient results from your panel code, XFOIL, and thin-airfoil theory. What do you notice about the results?

Figure (3) compares the lift coefficients predicted by the Vortex Panel Method (VPM), Thin Airfoil Theory (TAT), and XFOIL for the NACA 0015 airfoil. All three methods demonstrate approximately linear relationships between lift coefficient and angle of attack, with a zero-lift angle of approximately zero degrees, as expected for a symmetric airfoil.

The methods agree closely at small angles of attack but begin to diverge as the magnitude of the angle of attack increases. VPM predicts a greater lift-curve slope than TAT, potentially due to its consideration of the airfoil's finite thickness. XFOIL generally predicts lower lift coefficients than VPM at larger positive angles of attack, which may be attributed to viscous effects and boundary-layer development. Interestingly, XFOIL closely follows TAT throughout much of the investigated range.

Overall, the results demonstrate that all three methods provide similar lift predictions at small angles of attack, while differences in their aerodynamic assumptions become more apparent at larger angles.

Table 4: VPM and XFOIL Data for NACA 0015 with 200 nodes.

	VPM (Inviscid)			XFOIL ($Re = 3 \times 10^6$)		
α [deg]	$\tilde{C}_L$	$\tilde{C}_{m_{LE}}$	$\tilde{C}_{m_{c/4}}$	$\tilde{C}_L$	$\tilde{C}_{m_{LE}}$	$\tilde{C}_{m_{c/4}}$
-12	-1.47119	0.38094	0.02118	-1.31380	0.31807	-0.00390
-10	-1.22874	0.32032	0.01781	-1.10200	0.26809	-0.00370
-8	-0.98479	0.25815	0.01435	-0.87380	0.21104	-0.00560
-6	-0.73965	0.19472	0.01082	-0.66490	0.16311	-0.00240
-4	-0.49360	0.13034	0.00725	-0.44740	0.11089	-0.00080
-2	-0.24695	0.06533	0.00363	-0.22530	0.05624	-0.00010
0	0.00000	0.00000	0.00000	0.00010	-0.00003	0.00000
2	0.24695	-0.06533	-0.00363	0.22530	-0.05534	0.00100
4	0.49360	-0.13034	-0.00725	0.44740	-0.10369	0.00800
6	0.73965	-0.19472	-0.01082	0.66490	-0.16311	0.00240
8	0.98479	-0.25815	-0.01435	0.87380	-0.21104	0.00560
10	1.22874	-0.32032	-0.01781	1.10210	-0.26822	0.00360
12	1.47119	-0.38094	-0.02118	1.31380	-0.31807	0.00390
14	1.71184	-0.43969	-0.02444	1.45950	-0.33882	0.01620
16	1.95041	-0.49630	-0.02759	1.57970	-0.35327	0.02780

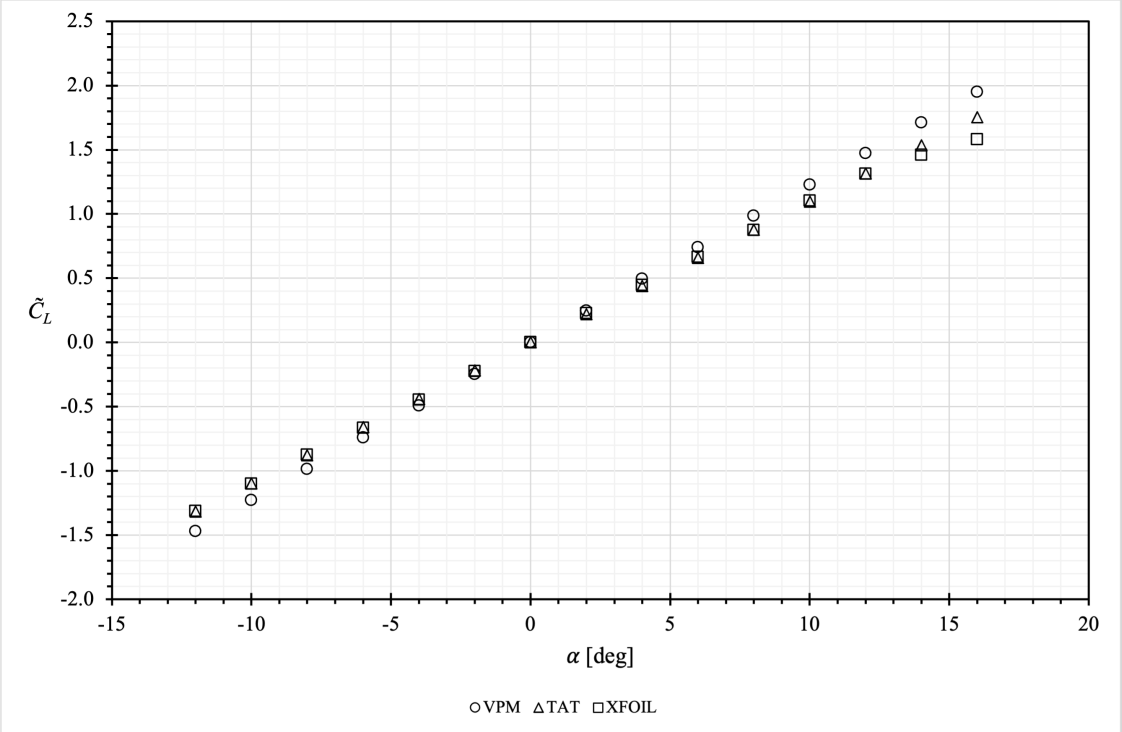


Figure 3: $\tilde{C}_L$ comparison between VPM, TAT, and XFOIL Data for NACA 0015.